

Java Database Connectivity (JDBC) with Microsoft Access Database using ucanaccess Driver

Aim

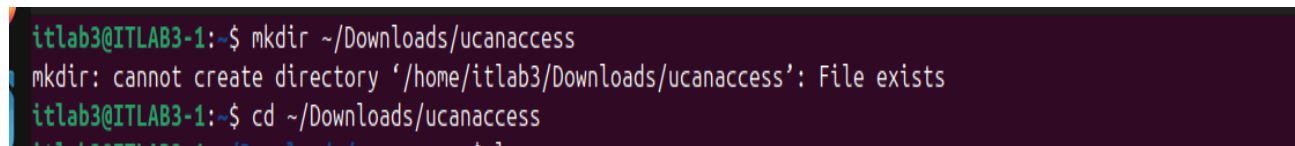
To connect a Java application to a Microsoft Access database using JDBC and retrieve records from a table.

Procedure / Steps

Step 1: Create a working directory

```
mkdir ~/Downloads/ucanaccess
cd ~/Downloads/ucanaccess
```

Screenshot



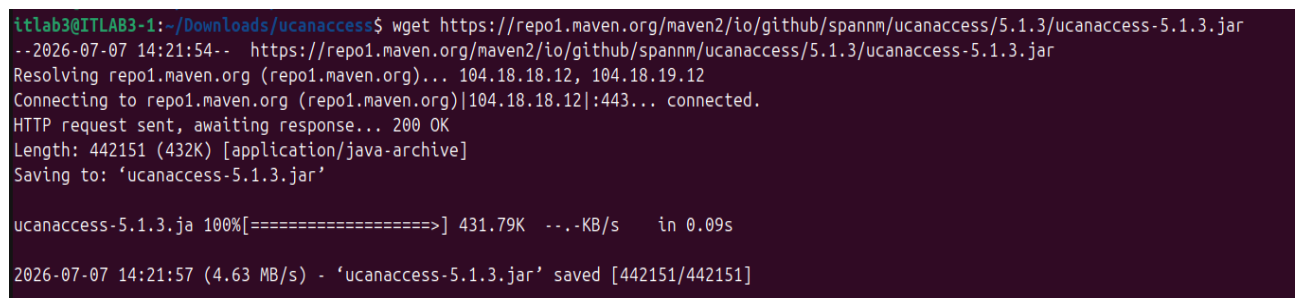
```
itlab3@ITLAB3-1:~$ mkdir ~/Downloads/ucanaccess
mkdir: cannot create directory '/home/itlab3/Downloads/ucanaccess': File exists
itlab3@ITLAB3-1:~$ cd ~/Downloads/ucanaccess
```

Step 2: Download the UCanAccess JDBC driver

```
wget
https://repo1.maven.org/maven2/io/github/spannm/ucanaccess/5.1.3/ucanaccess-5.1.3.jar
```

Download the required dependency files if using the non-uber version.

Screenshot



```
itlab3@ITLAB3-1:~/Downloads/ucanaccess$ wget https://repo1.maven.org/maven2/io/github/spannm/ucanaccess/5.1.3/ucanaccess-5.1.3.jar
--2026-07-07 14:21:54-- https://repo1.maven.org/maven2/io/github/spannm/ucanaccess/5.1.3/ucanaccess-5.1.3.jar
Resolving repo1.maven.org (repo1.maven.org)... 104.18.18.12, 104.18.19.12
Connecting to repo1.maven.org (repo1.maven.org)|104.18.18.12|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 442151 (432K) [application/java-archive]
Saving to: 'ucanaccess-5.1.3.jar'

ucanaccess-5.1.3.jar 100%[=====>] 431.79K --.-KB/s in 0.09s

2026-07-07 14:21:57 (4.63 MB/s) - 'ucanaccess-5.1.3.jar' saved [442151/442151]
```

Step 3: Copy the Microsoft Access database

Place the database file Database1.accdb inside the project directory.

Verify using:

```
ls
```

Expected files:

- AccessDemo.java

- Database1.accdb
- UCanAccess JAR file(s)

Step 4: Create the Java program

Create the source file.

```
nano AccessDemo.java
```

Write the Java JDBC program to connect with the Access database and fetch records.

Save the file.

Screenshot

```

GNU nano 7.2 AccessDemo.java
import java.sql.*;

public class AccessDemo {

    public static void main(String[] args) {

        try {

            String url =
                "jdbc:ucanaccess:///home/itlab3/Downloads/ucanaccess/Database1.accdb";

            Connection con =
                DriverManager.getConnection(url);

            System.out.println("Connected Successfully");

            Statement st =
                con.createStatement();

            ResultSet rs =
                st.executeQuery("SELECT * FROM student");

            while(rs.next()) {

                System.out.println(
                    rs.getInt( "ID" )+ " " +
                    rs.getString( "NAME" )+ " " +
                    rs.getInt( "MARKS" ));

            }

            con.close();

        } catch (Exception e) {

            e.printStackTrace();

        }

    }

}
  
```

Read 42 lines

Help Write Out Where Is Cut Execute Location M-U Undo M-A Set Mark M-J To Bracket M-O Previous ^B Back
 Exit Read File Replace Paste Justify Go To Line M-E Redo M-G Copy ^O Where Was M-W Next ^F Forward

Step 5: Compile the program

```
javac AccessDemo.java
```

If there are no errors, AccessDemo.class will be created.

Result: AccessDemo.class Created

Step 6: Execute the program

Run the Java program using the JDBC driver.

```
java -cp ".:ucanaccess-5.1.2-uber.jar" AccessDemo
```

The program establishes the database connection and displays the records stored in the Access database.

Screenshot

```
itlab3@ITLAB3-1:~/Downloads/ucanaccess$ javac AccessDemo.java
itlab3@ITLAB3-1:~/Downloads/ucanaccess$ java -cp ".:ucanaccess-5.1.2-uber.jar" AccessDemo
Connected Successfully
1 RAMESH 10
2 MAHESH 20
3 SURESH 30
4 RAJESH 40
```

Program Logic

1. Import JDBC packages.
2. Load the UCanAccess JDBC driver.
3. Specify the path of the Access database.
4. Establish the database connection.
5. Execute the SQL query.
6. Retrieve records using a ResultSet.
7. Display the retrieved records.
8. Close all database resources.

Sample Output

Connected Successfully

```
1 RAMESH 10
2 MAHESH 20
3 SURESH 30
4 RAJESH 40
```

Result

The Java program successfully connected to the Microsoft Access database using the UCanAccess JDBC driver and retrieved all records from the database table. The output was displayed successfully on the terminal.